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Enhancing RAG Comprehension of Complex Reference Tables in Technical Documents

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Abstract. In RAG-based question-answering systems, built on corpora containing technical documents, the presence of reference tables poses a unique challenge. Unlike the continuous text on which LLMs and embedding models are primarily trained, reference tables require attention to two dimensions. To effectively answer questions based on these structures, a combination of spatial, linguistic and numerical reasoning is required. Although work has been done to assess and improve the ability of RAG systems to retrieve non-textual segments from document corpora, the scope is often wide, targeting images, graphs, diagrams and tables as a whole. Also, many works focus only on retrieval, and the task of ultimately answering the question is left unaddressed. In this paper, we will assess the ability of a RAG system to answer both text and table based questions. We will demonstrate the degraded performance that standard implementations exhibit on tabular data and explore and evaluate methods aimed at improving this performance. By combining targeted pre- and post-retrieval techniques we will ultimately bridge the text-table performance gap achieving an accuracy improvement from 48.9% to 84.4% on table-based questions whilst maintaining performance on text-based questions.

Keywords: RAG · Multi Modal RAG · Reference Tables · VLMs · Chain-of-Thought

1 Introduction & Background

The rapid adoption of Large Language Model (LLM) based question-answering systems across organisations represents a fundamental shift in how institutional knowledge is accessed and utilised. Retrieval Augmented Generation (RAG) [11] has emerged as the dominant paradigm for grounding AI systems in organisational data while circumventing the limitations imposed by context windows. This approach enables organisations to deploy sophisticated *chatbots* that can query internal knowledge bases using natural language, transforming how employees, clients, and stakeholders interact with institutional information.

RAG systems rely on the principle of semantic similarity between query embeddings and document chunk embeddings, typically measured through cosine similarity in high-dimensional vector spaces [9]. The architecture involves preprocessing document corpora into manageable chunks, generating dense vector representations using an embedding model, and storing these in a vector database for efficient retrieval. During inference, user queries are embedded into the same vector space, and the most semantically similar chunks are retrieved and provided as context to the generation model. This approach has proven highly effective for knowledge-intensive tasks and has significantly reduced the hallucination problem that plagued early LLM deployments [11].

However, the assumption underlying RAG systems, that semantic similarity in continuous text embeddings adequately captures relevance, is called into question when applied to structured data formats, particularly reference tables. Unlike the predominantly textual corpora on which embedding models are trained, tabular data requires simultaneous attention to spatial positioning, numerical relationships, and linguistic content for effective retrieval. The linear nature of text embeddings struggles to adequately capture the two-dimensional semantic relationships that are crucial for identifying relevant tabular content based on user queries.

When processed as symbols, reference tables raise further questions at the generation phase of the RAG process. Tables encode information through two-dimensional layouts where row-column intersections carry semantic meaning that cannot be preserved through linear text serialisation. This spatial structure presents a critical challenge for language models, as LLMs process input by converting visual or structured information into sequential token streams that obscure the positional relationships essential for table comprehension. This limitation becomes particularly problematic in technical documents, where essential information is frequently encoded in complex tabular formats that require spatial reasoning to accurately interpret.

The challenge is further compounded by the fact that standard document processing pipelines, optimised for continuous prose, often fragment tables during the chunking process. Traditional character-based or token-based chunking algorithms have no awareness of table boundaries, frequently splitting tables across multiple

chunks and destroying the spatial relationships that are crucial for comprehension. By taking a more table-aware approach to structuring text and ensuring tables are not split across multiple chunks it may be possible to alleviate some of the text-table performance gap. However, as this research will show, even when tables remain intact within chunks, and they are structured appropriately, the one-dimensional encoding of text in LLMs strips away the visual structure that enables two-dimensional reasoning.

The practical impact of poor table handling in RAG systems extends beyond academic interest. In domains such as regulatory compliance, financial analysis, and scientific research, critical information is frequently encoded in tables. A RAG system that cannot accurately retrieve and reason over tabular content represents a significant limitation for organisations seeking to democratise access to their technical documentation. RAG implementations across many industries are currently in the process of transitioning from experimental to business critical, underscoring the importance of carefully evaluating and addressing the fundamental limitations that could undermine system reliability. The trust that users place in RAG-powered systems, stemming from their apparent grounding in authoritative sources, lends particular urgency to this research.

Current evaluation frameworks for RAG systems predominantly utilise benchmark datasets derived from textual knowledge bases such as Wikipedia and other internet-based corpora. TriviaQA, HotpotQA and PopQA are just a few datasets that have risen to prominence, being adopted by many RAG research papers [6]. While these benchmarks effectively evaluate RAG performance on continuous text, they fail to capture the unique challenges posed by structured data. This evaluation gap has resulted in a systematic underestimation of performance degradation on real-world document corpora containing mixed content types.

The multimodal capabilities of recent embedding models and vision-language models present potential solutions to the table processing challenge. Models capable of processing both text and images can handle tables as visual objects, preserving their spatial structure while enabling semantic retrieval. However, the integration of multimodal capabilities into RAG systems introduces new complexities, particularly around the modality gap; the tendency for image and text embeddings to cluster separately in a vector space, potentially biasing retrieval toward one modality over another [13].

Recent work has begun to address these challenges through various approaches. Some researchers have explored reducing the modality gap through further training [7], while others have investigated specialised embedding models for handling diverse content types [17]. Vision-language models have demonstrated capabilities in understanding complex document layouts and visual elements [25], and there has been growing interest in hybrid retrieval systems that process multiple modalities simultaneously [2].

However, most existing work approaches visual content as a monolithic category, treating images, diagrams, charts, and tables with uniform methodologies. While this broad approach addresses general multimodal challenges, it fails to focus on the unique issues that intersecting spatial, linguistic and numeric reasoning pose in the form of tables. At the very least, the broad approach to multimodal RAG obfuscates the performance of tables into a broad result. Furthermore, most existing work focuses on improving retrieval accuracy in isolation, focusing on Query-page pair benchmarks such as ViDoRe [4] and presenting their final findings using nDCG@5 metrics.

This research gap not only presents an opportunity for our work, but also a challenge. A robust QA dataset, of Question-Answer pairs, is required to evaluate a RAG system’s ability to answer questions from reference tables. The unique and focussed aim of this research encounters an initial hurdle: the absence of existing datasets specifically designed for this task. Many datasets covered aspects relating to our objective but had key deficiencies that made them unsuitable for this project.

Of the available datasets, several had fundamental limitations in our practically oriented, table focussed, RAG context. DocVQA [15], assumed visual elements were already present alongside the question, with queries like *"What date is seen at the top of the letter?"* that do not translate effectively to a RAG framework where the letter in question must be returned based on the semantic similarity to the query. InfoVQA [16] contained an insufficient number of tables for comprehensive evaluation. Other datasets, such as TabFQUAD [19], consisted primarily of captioned tables, making them potentially suitable for retrieval tasks but inadequate for question answering evaluation. Additionally, some datasets included multiple-choice questions [12], which are inappropriate for RAG QA contexts where users do not have predetermined answer options. We acknowl-

edge that TAT-QA [29] could have served as a viable alternative to our approach, though it was discovered too late in the project timeline to incorporate effectively.

A central aspect of this project is addressing the table-text gap from the perspective of an organisation implementing RAG systems with fresh document corpora. Therefore, after reviewing the available datasets, it seemed apt to create a new dataset derived from a raw set of documents, rather than working with pre-curated datasets of text chunks and tables. To produce a high-quality and varied dataset we will collate a raw document corpus of at least 500 pages and create 150 QA pairs from their contents.

Our objective of end-to-end RAG evaluation opens up opportunities to experiment with post-retrieval techniques. We reviewed the available reasoning enhancement strategies for LLMs, considering several prominent methods. Tree-of-Thoughts [24] explores multiple reasoning paths simultaneously but introduces complexity and computational overhead unsuitable for production RAG systems. By contrast, Plan-and-Solve [21] took a zero-shot, generalised approach to problem decomposition that did not align with our focused scope where concrete examples are more instructive. Self-Consistency [22] demands multiple inference passes that significantly increase latency and cost. Arguably the direct inference steps taken in reference table comprehension makes a greedy approach to reasoning more appropriate.

Chain-of-Thought (CoT) prompting [23] emerged as the preferred approach for our table-focused RAG framework due to its direct applicability to spatial reasoning tasks and seamless integration across different modalities. Unlike alternatives requiring specialised infrastructure or multiple inference steps, CoT’s implementation at the prompt level enables uniform application across text-based, image-based, and hybrid retrieval methods while maintaining single-pass efficiency. Most critically, the sequential nature of CoT reasoning — identifying relevant tables, locating specific rows and columns, and extracting target values — naturally aligns with the cognitive processes required for table comprehension, making it particularly well-suited for the spatial and numerical reasoning demands of tabular question-answering tasks. Multi-modal CoT has also recently been introduced [26] where reasoning steps are presented in multiple modalities.

In this work, we will address the table-text performance gap through a comprehensive evaluation of retrieval and generation techniques specifically designed for mixed-content document corpora. Unlike previous studies that focus primarily on retrieval accuracy or treat multimodal content as a monolith, we evaluate end-to-end question-answering performance on a novel QA dataset derived from a realistic document corpus containing both textual and tabular information. We systematically compare text-based, image-based, and hybrid multi-modal approaches, providing insights into the trade-offs between different architectural choices for handling structured content in RAG systems. Our contributions can be summarised as follows:

1. To the best of our knowledge, we are the first to take a table-focused approach to RAG evaluation from an end-to-end perspective.
2. A novel RAG dataset designed to test the ability of implementations to answer targeted questions about tabular content.
3. A specialised, practical framework designed for handling complex document corpora and answering table based questions in a RAG context.

2 Methodology

In this section we will discuss the creation of our novel dataset from a raw document corpus and describe in detail the implementation of each technique we evaluated in this research. Finally, we will describe the evaluation framework with which we assessed each of these methods, explaining which metrics were used.

2.1 Dataset Creation

Following extensive exploration, we identified the Water Framework Directive (WFD) [3] as an excellent source of suitable documents for our QA dataset. The WFD is an EU directive focused on protecting and improving water environments. Their website publicly hosts a comprehensive corpus of guidance documents that explain methodologies for achieving good water status under various conditions. The corpus contains 38 PDF documents with 3888 total pages. Given our project scope required approximately 500 pages, we narrowed the

selection to three documents: *Environmental standards phase 2_Final_110309.pdf*, *Guidance No 01 - Economics - WATECO (WG 2.6).pdf*, and *Guidance No 27 - Deriving Environmental Quality Standards - version 2018-1.pdf*. These documents were selected specifically for their high concentration of complex reference tables, with a combined page count of 568.

A semi-synthetic QA example generation approach was employed to create our final dataset. This methodology involved developing a specialised utility that processed the selected documents page-by-page, feeding the content to an LLM. The LLM received prompts designed to generate QA pairs where questions were answerable using page content and answers were concise and specific (constrained to 1-5 words). See the Appendix for all prompt templates. Each generated QA pair was subsequently reviewed and could be manually accepted, edited, or discarded. This hybrid system significantly accelerated question creation while the manual review process served as a quality control mechanism to minimize potential bias. We took particular care to ensure that table-based questions could not be answered using other textual content within the documents, maintaining the integrity of our table-specific evaluation approach.

It was also important that any questions we created could not be answered through general knowledge or contextual inference. Each question must require direct access to and accurate interpretation of the structured tabular information. This characteristic makes our dataset particularly valuable for evaluating the true table-processing capabilities of RAG systems, as opposed to their ability to leverage background knowledge or textual context clues. To ensure this, each question was asked of the same LLM without the originating context; if the LLM gave the correct answer - showing that the information was present in its training data - the question was discarded.

Table 10: Geological types for lakes					
1 st tier types	Type	Catchment	Alkalinity	Conductivity ¹	Colour
(Geology)			mg/l of calcium carbonate	μS/cm	mgPt/l
Siliceous	Low Alkalinity	More than 90 per cent siliceous solid geology	< 10	< 70	<= 30
	Moderate Alkalinity	More than 50 per cent siliceous solid geology	10 to 50	70 to 250	
Calcareous	High Alkalinity	More than 50 per cent calcareous geology	> 50	> 250 to 1000	
	Marl	More than 65 per cent limestone			

Fig. 1. Example table showing water quality thresholds for different lake types

To illustrate the nature of our dataset, consider the following question with relation to Fig. 1: "What is the reference threshold of Colour for a Moderate Alkalinity Siliceous Lake?". This question exemplifies the specific, technical queries that characterise our evaluation framework. The answer must be extracted directly from the tabular data shown in Figure 1 and cannot be found elsewhere in the document; it demonstrates the precision required for effective table-based question answering in specialised domains.

This particular example highlights several key challenges inherent in table-based QA systems. First, the question requires precise navigation through multiple table dimensions (lake type, alkalinity level, and colour). Second, the answer demands exact extraction of numerical values with appropriate units, where approximate responses would be inadequate for practical applications. Finally, the value in question spans multiple rows in the table, requiring that the LLM not only understands the table dimensions but also can follow the delineation of cells in the context of others.

Care was taken to require equivalent specificity from the text questions. For example figure 2 shows a section of text from *Guidance No 01 - Economics - WATECO (WG 2.6).pdf*. The question for this segment is "When was the Water Framework Directive published in the Official Journal?" expecting the answer to be:

"December 22, 2000". With this level of clarity in the questions, we aim to fairly evaluate the ability of our RAG framework to extract information from text and tables.

December 22, 2000, will remain a milestone in the history of water policies in Europe: on that date, the [Water Framework Directive](#) (or the Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 establishing a framework for Community action in the field of water policy) was published in the Official Journal of the European Communities and thereby entered into force!

Fig. 2. Example text from environmental standards document showing publishing milestone

Through this process, we constructed a dataset containing 150 questions, with equal distribution between table-based and text-based answers (75 questions each). This balanced approach enables comprehensive evaluation of system performance across both content types. Although all questions are contained within the same dataset file, they are tagged with their relevant content type for separation of results. As such, we will refer to them separately as TextQA and TableQA. We also ensured that next to each question was recorded the page from which it was created to be able to assess our frameworks on retrieval as well as correctness.

For all our work we made use of Git for source control. As such, both the dataset and the utility developed for its creation are publicly available on this project’s GitHub repository [1]. We actively encourage other researchers to contribute additional examples to expand this resource. Having created our question set comprised of TextQA and TableQA pairs, we needed to establish a baseline result upon which to improve.

2.2 Baseline

To set this baseline figure we took the WFD PDFs and extracted them into text using PyPDF (a standard PDF text extraction library for Python). Following a typical approach [11], we segmented the text into 1000-character chunks using LangChain’s recursive text splitting functionality, with a 200-character overlap to preserve context at chunk boundaries. Each chunk was then processed through an embedding model to generate dense vector representations, which were stored in a Chroma vector database alongside their respective text chunks. For this baseline implementation we employed the well used text-ada-embedding-002 model (OpenAI, 2022). While no peer-reviewed publication details this model’s architecture or training methodology, it represents the state-of-the-art commercial embedding model available at the time of our research and a popular model with organisations due to its accessibility through APIs.

Using this vector store, we implemented a standard RAG pipeline. For each question-answer pair, the question was encoded using the same embedding model, and cosine similarity scores were calculated between the question embedding and all stored chunk embeddings. The content from the top 4 most similar chunks was then retrieved and passed to the generation model alongside the question (we selected 4 chunks to maintain consistency with other methods in our comparison). For generation model we selected GPT-4o (OpenAI, 2024) as it represents an accessible, state-of-the-art, multi-modal LLM. Following Liu et al. [14] who demonstrated that extreme positions are most impactful for relevant chunks, we positioned the highest-scoring chunk nearest to the question. However, as per Jin et al. [8] at this context size, any more elaborate reordering would yield negligible improvements.

This baseline approach achieved 48.9% correctness on table-based questions compared to 73.2% correctness on text-based questions (see Table 1), as evaluated using an LLM-as-a-judge methodology [27], which we will discuss in detail later on. Figure 3 illustrates the fundamental limitation of this naive approach, showing how much of Figure 1’s table structure is lost during text extraction. Even a human expert would struggle to answer the example question given this degraded context, which likely explains the substantial performance gap between text and table questions. To address this limitation, we next explore more structure-aware approaches.

```

Alkalinity Conductivity1 Colour 1st tier types
(Geology) Type Catchment
mg/l of calcium
carbonate  µS/cm mgPt/l
Low Alkalinity More than 90 per cent
siliceous solid geology < 10 < 70
Siliceous Moderate
Alkalinity More than 50 per cent siliceous solid
geology 10 to 50 70 to 250
High Alkalinity More than 50 per cent calcareous geology Calcareous
Marl More than 65 per cent
limestone > 50 > 250 to 1000 <= 30

```

Fig. 3. Text Extraction of Figure 1

2.3 Structured Text

The performance degradation observed with table-based questions in our baseline approach clearly demonstrates the limitations of naive text extraction methods. The first, and most straightforward, approach we will explore to address this issue is to use structured text extraction that preserves the tabular formatting essential for accurate comprehension and retrieval.

Our structured approach applies two distinct modifications to the baseline methodology. First, we replace standard PDF text extraction with a markdown-based extraction process using `pypdf4llm`, a Python library specifically designed to maintain tabular formatting during PDF processing. This approach converts PDF tables into markdown format. Unlike plain text extraction, which loses tabular structure, markdown makes use of additional characters like pipes (|) and dashes (-) to preserve column and row relationships.

Second, we modify our chunking strategy from character-based segmentation to page-based chunking. This change directly addresses the table fragmentation problem identified in the introduction, where traditional chunking methods can split tables across multiple chunks, destroying their semantic coherence. Since our analysis of the WFD documents reveals that the majority of tables are contained within single pages, page-based chunking significantly reduces the probability of table segmentation while maintaining manageable chunk sizes for the embedding model. For comparison we will implement the per-page design for unstructured text also.

It should be noted that this page-based approach introduces some constraints to the RAG framework's flexibility. The method is inherently dependent on document formats with explicit page boundaries and may be less suitable for documents where content naturally spans multiple pages or where page breaks do not align with logical content divisions. However, for structured documents like the WFD corpus, where tables are typically self-contained within pages, this trade-off between flexibility and structural preservation appears justified.

Although the structured text and page-based chunking approach provides a valuable comparison point in our evaluation, this solution retains fundamental limitations that constrain its potential effectiveness. First, the approach remains dependent on the reliability of markdown extraction, which is rarely entirely accurate on more complex tables. More critically, however, is the inherent architectural limitation of text-based language models: regardless of whether text is structured or unstructured, LLMs process input by serialising characters into one-dimensional token sequences [20]. While the remarkable effectiveness of transformer architectures allows some table-based questions to be correctly answered, this constraint remains; that LLMs attend to text information in a single dimension. Consequently, the spatial relationships, visual hierarchies, and two-dimensional layout properties that are integral to table comprehension cannot be fully captured through text-based representations alone. This limitation suggests that only vision-based models capable of processing spatial information can truly comprehend the complete semantic structure of tabular data in their original format.

2.4 Images

Processing the tables in our raw document corpus as images introduces some additional considerations and complexities. RAG systems are predicated on a fair scoring mechanism to rank chunks based on their affinity

with the query. On a text-to-text basis this scoring mechanism is often a simple cosine similarity score as described in section 2.2. With images as part of the chunked content we must change this approach. One possibility is to switch to a joint image and text embedding model, keeping all else equal. However, as others have shown [13], the modality gap produced by training a joint image and text embedding model is significant. In a RAG context, this gap inevitably creates performance issues, as text-based queries will systematically exhibit higher affinity scores with text-based chunks than with image-based chunks, introducing retrieval bias.

With the assumption that achieving truly fair cross-modal similarity measurements remains beyond the current state-of-the-art, we identify three viable approaches for incorporating images into our RAG framework while mitigating modality gap issues. We will outline the implementation of each of these in the following sections.

Implementation 1: Captioning. Following an approach similar to those employed by Jiang et al. [7] and Riedler et al. [18], we instruct an LLM to generate descriptive captions for table images. These captions serve as the embedding text for similarity scoring, maintaining compatibility with text-based retrieval mechanisms. During context compilation, the original images replace their captions in the final prompt sent to the generation model, preserving visual information for the answering phase.

This required an initial preprocessing step using OCR and Computer Vision to extract the tables and text from the PDF pages. Using OpenCV, we created bounding boxes around the tables, capturing those sections as images and then using GPT-4o for character recognition of the remaining page (see Fig. 4). Using GPT-4o resulted in almost perfect text extraction when compared with the true text. Extracting these into a well-structured file system made them useable across other techniques.

The text data, output from the OCR process, were combined into a single continuous string which was chunked and processed in a manner identical to the baseline implementation in section 2.2. The images resulting from the process outlined in Figure 4 were then passed to GPT-4o with a prompt requesting a description of the table. This description was then embedded and stored in the same vector store as the text chunks. Alongside the embedding was stored the base 64 encoded representation of the table image which was passed to the generation model when that embedding was returned as one of the top 4 scoring.

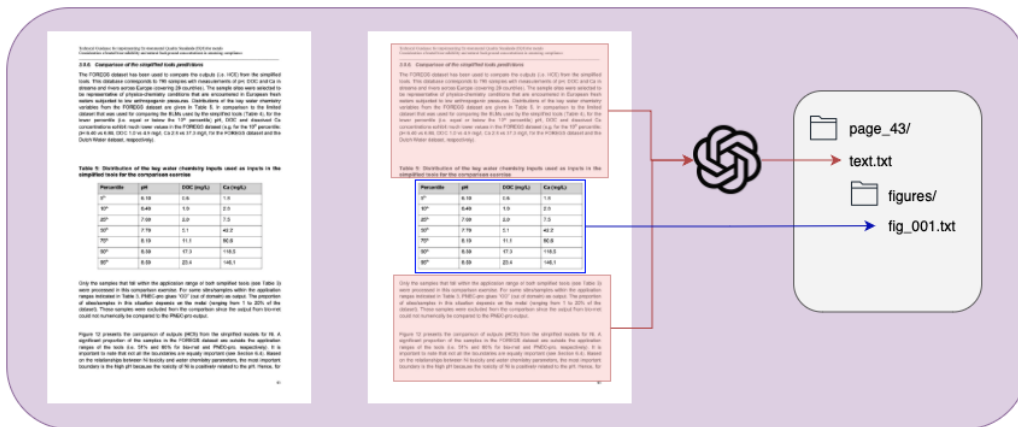


Fig. 4. OpenCV and OCR Page Extraction

Implementation 2: Pure Image Retrieval. This approach eliminates text entirely by processing each document page as a complete image and storing these in an image-based vector store. Since all stored vectors originate from images, text-to-image bias is avoided while maintaining the ability to score image embeddings against text queries. We used the ColPali [4] model for this implementation, as among available models capable of this cross-modal embedding this seemed the most suitable, having been specifically fine-tuned for PDF document retrieval tasks.

The size of the ColPali model required the use of a cloud GPU in the form of Google Colab's T4. We loaded the pre-trained model weights into the environment and built a custom API to handle inference requests.

Building this interface was a challenging aspect of this process. To use the custom embedding model we also needed to design a bespoke vector store interface using LanceDB, a python vector store library. Both the API and the vector store are available on our GitHub repo for others to use.

With the established infrastructure and vector store, document processing becomes straightforward. Rather than employing text chunking strategies, each PDF page is treated as an individual image and embedded and stored as such. During inference, textual queries are embedded using the same multimodal encoder, leveraging the model’s capacity for cross-modal representation learning. Crucially, since the vector store exclusively contains dense embeddings of images, the modality gap is mitigated.

Implementation 3: Dual Vector Store Architecture. This approach circumvents joint embedding limitations by maintaining separate vector stores for text and image content. By retrieving distinct document sets from each modality and combining them during context compilation, we eliminate cross-modal scoring issues while preserving system flexibility and avoiding the constraints of either pure-text or pure-image approaches.

Our novel dual architecture framework (see Figure 6) leverages the same preprocessed data extracted through the OpenCV and OCR pipeline described in Figure 4. The separated text content, now free of embedded tables, undergoes conventional chunking procedures optimised for continuous plain text. This separation ensures that text chunking operates under ideal conditions without interference from the tables.

Concurrently, extracted table images are processed into a dedicated image vector store, maintaining complete separation between modalities throughout the embedding and storage phases. This architectural decision eliminates the need for cross-modal similarity comparisons during retrieval. At the retrieval phase, the top 2 chunks/images from each store are included in the context to the model. This is to ensure fairness with the other methods where the top 4 is used.

For this implementation, we employed ColPali as the image embedding model, drawing motivation from its demonstrated effectiveness on table-focused datasets such as TabFQUAD in the original paper [4]. While acknowledging that a model specifically fine-tuned for table image embeddings might yield superior performance, ColPali’s broader document-oriented training offers practical advantages for systems requiring diverse image type support within a unified framework. This versatility enables the embedding of various visual elements within a single vector store, enhancing system scalability. Also, using the same model as our pure-image implementation serves as a more direct comparison between pure-image and hybrid in the results of this research.

2.5 Chain-of-Thought Reasoning

The distinctive contribution of our research lies in adopting an end-to-end evaluation approach for RAG systems containing documents with tables, assessing not merely retrieval accuracy but the complete framework’s capacity to generate correct answers from corpus-specific information. While retrieval performance constitutes a critical bottleneck, successful information location represents only the initial phase of the reasoning process. Once the appropriate chunk, page, or image is retrieved, the generation model must demonstrate reasoning capabilities over the provided context.

Following a review of techniques for eliciting reasoning in LLMs, Chain-of-Thought (CoT) prompting emerged as the most versatile approach for our experimental framework. CoT’s implementation at the context compilation stage, through the incorporation of reasoning examples into the prompt [23], enables uniform application across all retrieval methodologies, facilitating direct performance comparisons when tables are presented across different modalities.

Consider your own thought process when answering a question based on a reference table. For tabular information a variety of reasoning is required. First, the relevant table must be selected from the context. Then, the correct row and/or column should be identified. Next, based on the nature of the question (aggregation, single value extraction, category identification, etc) a referencing step is considered where the cell(s) required to answer the question are located by reading across a row or up a column. Finally, the resulting answer can be calculated.

As an example take this question with Fig. 5 as the context: *"Where were the highest number of Moderate water bodies classified with 95% confidence?"*. First, you would need to look at the column headings and

identify the *Moderate* column in the *95 per cent confidence* tranche, leading you to the end column (column 6). Then, you would find the largest number in that column giving you 29 on row 1. Next, you would read across to the regions in the row labels and identify the correct answer: *England*. We are using this kind of task decomposition to prompt the generation model, following Zhou et al. [28].

Table 21: Classification of water bodies with sufficient data						
	Face value ¹			95 per cent confidence ²		
	High	Good	Moderate	High	Good	Moderate
England	3	6	34	7	7	29
Northern Ireland	3	2	5	6	2	2
Scotland	42	9	0	47	4	0
Wales	6	6	5	9	4	4

Fig. 5. Impact Ratings Example

For the image-only or multi-modal RAG implementations, we adopt the extended framework proposed by Zhang et al. [26], who demonstrated significant improvements in reasoning quality through the integration of visual inputs with the reasoning prompt and with an explicit rationale generation step. By including the image of the table in the prompt we can connect the reasoning more concretely to the context and by generating a rationale step we encourage the model to make a plan of how it will achieve the result.

In Figure 6 we demonstrate the novel dual-vector-store architecture described earlier in the methodology combined with the Chain-of-Thought prompting additions. The combination of all these approaches exemplifies our targeted approach to table QA. The design for Figure 6 was inspired by a similar diagram by Faysse et al. [4] although the process described and elements within are distinct.

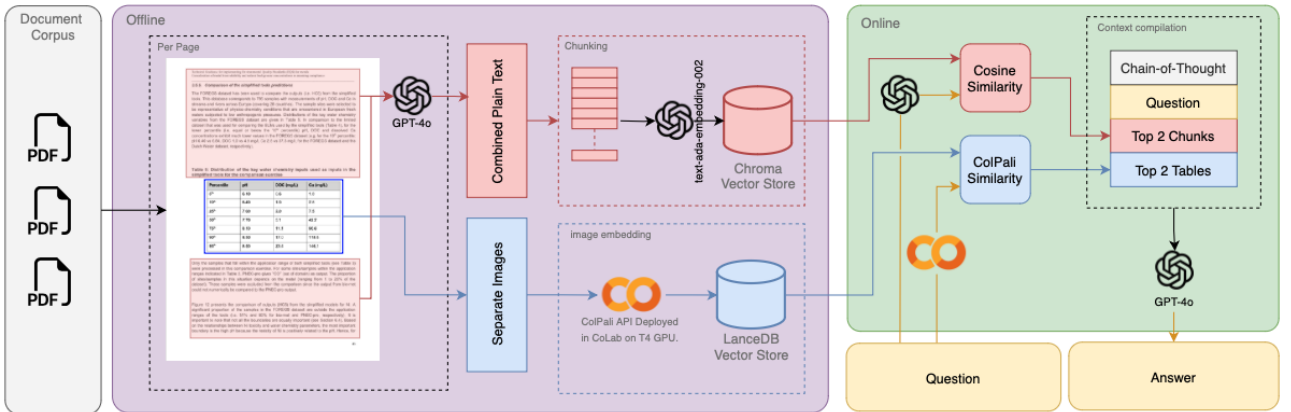


Fig. 6. Full Dual Vector Store Framework

2.6 Evaluation Framework

To evaluate and compare all of the implementations described in this section, we designed a systematic testing process that evaluates each method individually on our dataset of question-answer pairs. The evaluation process passes each question through the respective framework and captures the final answer. Importantly, all methods use identical prompts for the final generation step, with CoT and Multi-modal CoT (MM CoT) prompt additions maintained consistently across all relevant implementations.

Along with the generated answer, each implementation returns metadata indicating the source documents: image metadata contains page and figure numbers for extracted tables, while page-based implementations return the originating page number. For the chunk-based methods, different implementations required distinct pre-processing approaches (plain text, structured text, OCR), therefore maintaining consistent chunk identifiers across the dataset proved impractical. While this could be viewed as a limitation, as pre-prepared chunks with standardised IDs would enable more comprehensive evaluation, this approach provided greater implementation flexibility. Consequently, only implementations using full-page or table-image vector stores support document retrieval scoring.

As discussed previously, we perform no reordering of retrieved results, enabling a straightforward binary document retrieval accuracy calculation per question-answer pair: whether the correct page or table appears in the returned documents. We supplement this with ROUGE-1 scores [5], measuring 1-gram overlap between generated and ground truth answers. This established metric complements the more modern LLM-as-a-judge approach which we will be using as our primary evaluation metric.

Following Zheng et al. [27] we use LLM-as-a-judge to score Correctness. To calculate this metric, the question, the true answer and the expected answer are passed to a final judge LLM which is prompted to return "Y" or "N" based on whether the answer is correct or not. When compared with human evaluation this approach resulted in almost identical scores giving us confidence that we can take it as truthful. We made sure to not use the same model for generation and judgement to prevent bias. For the judgement model we used GPT-4.1, the most recent model released by OpenAI at the time of writing.

3 Experimental Results and Discussion

Following the methodology described in the previous section, we conducted a comprehensive study of the aforementioned techniques. The results of each experiment are shown in Table 1. The table shows the Correctness score, evaluated using LLM-as-a-judge, ROUGE1 and Retrieval Accuracy calculated by matching the page or figure id if present in the returned documents. Correctness and Retrieval Accuracy are each presented as a percentage. Since each individual result is a binary, the plain text interpretation of these scores are: the percentage of questions for which the final answer was correct and the percentage of questions for which the correct document was retrieved, respectively. ROUGE1 has been scaled up to a score out of 100 to make comparison with Correctness more natural. Retrieval Accuracy scores are omitted for CoT variants as they remain identical to their non-CoT counterparts. A dash (-) indicates a score was not calculated. Methods are presented in the order they were discussed in the methodology section.

As you can see from the results, the correctness score tracks fairly closely with ROUGE1 while being more forgiving on text questions where answers that are semantically identical would still qualify as correct even if the wording of the answer had less overlap. Based on example-by-example analysis, the LLM-judged correctness corresponds more closely with human evaluations, so we will mostly focus on this metric in our analyses.

The first result in our table justifies our work on this problem. Performance of a baseline RAG system on table-based questions is clearly degraded, showing a significant 24.4 percentage point difference when compared with equivalent text-based questions.

These results demonstrate many interesting things about the interaction of text and image embeddings in the context of a multi-modal RAG framework and how multi-modal compares with pure text and pure image based RAG systems for correctness on table and plain text questions. The standout performances of the ColPali-based implementations is clear from these results, with the per-page image-only method exhibiting the best overall performance on our stated objective: improving correctness on table based questions without degrading text performance. In fact, the combination of the ColPali embedding and similarity scoring model along with multi-modal chain-of-thought prompting produced a 35.5 percentage point improvement over the naive baseline implementation on the TableQA questions whilst maintaining and even slightly improving the text based score by 5.4 percentage points.

Along with this standout result, our experiments demonstrate other subtle effects of each technique. We see that, as well as improving readability to the human eye, structuring text as markdown does produce a fair

Table 1. Experimental Results: Performance Comparison Across Different Methods

Method	Correctness(%)		ROUGE1 (Avg.)		Retrieval Acc. (%)	
	TextQA	TableQA	TextQA	TableQA	TextQA	TableQA
Chunked Text (Baseline)	73.2	48.9	61.7	42.6	-	-
+CoT	69.6	53.3	62.7	50.6	-	-
Chunked Markdown	67.9	55.6	60.3	50.4	-	-
+CoT	67.9	55.5	60.8	55.9	-	-
Per-Page Text	67.9	60.0	62.1	55.9	62.5	82.2
+CoT	67.9	71.1	61.3	66.3		
Per-Page Markdown	69.6	68.9	66.7	63.0	66.1	80.0
+CoT	69.6	73.3	64.2	69.0		
Table Image Captioning	77.3	62.2	70.6	56.8	-	66.2
+CoT	76.0	64.4	69.9	55.4		
Per-Page ColPali	76.7	80.0	72.5	76.1	83.9	91.1
+CoT	71.4	77.8	67.8	79.7		
+MM CoT	78.6	84.4	71.1	81.4		
Dual Text-Image Framework	80.4	77.8	72.3	75.4	-	89.9
+CoT	78.6	75.6	71.6	74.6		
+MM CoT	73.2	73.3	70.3	73.5		

Note: Results across 75 TextQA and 75 TableQA questions

improvement (+6.7 pp) on TableQA over the baseline in the chunked context. However, performance of this technique in comparison to other results in the table is still low. This is likely due to the markdown content being flattened to a continuous string when consumed by the LLM, resulting in the degradation of 2-dimensional spatial reasoning. Furthermore, the addition of other symbols (l, *, etc) introduced when parsing text into markdown is potentially influencing the embedding vectors of those chunks, resulting in slightly degraded retrieval, and therefore overall (-5.3 pp), performance on TextQA.

The per-page experiments for plain text and markdown provide additional insight, showing an 11.1 percentage point and 13.3 percentage point improvement on TableQA over their respective non-CoT chunking counterparts. This improvement we can attribute to the per-page method removing the risk of a table being split in half during chunking as the tables in our dataset typically only span a single page.

Another interesting result can be observed when comparing both the per-page and chunking versions of the text and markdown methods. We see that Chain-of-Thought prompting largely closes the performance gap between text and markdown, with only a 2.2 percentage point difference on table questions between CoT markdown and CoT text in both the per-page and chunked cases. This suggests that, for table questions, CoT largely makes up for not structuring the plain text as markdown.

The table captioning multi-modal RAG method demonstrates a fair improvement over both text and markdown chunking methods for table questions, with a 13.3 and 6.6 percentage point improvement respectively. This coincides with a stronger performance on the text-based questions over the baseline (77.3 pp vs 73.2 pp). This is most likely a result of the cleansing effect that the OCR process had on some of the document pages. The removal of the table information altogether from the text chunks likely makes them more suitable to RAG retrieval resulting in better scores on the chunks affected.

The experiment using dual image and text vector stores with ColPali as the image embedding model scored exceptionally well on both the TableQA and TextQA datasets. Although lagging behind pure image ColPali by 2.2 percentage points without CoT and actually exhibiting worse performance with CoT, this method produced the best performance of all the techniques on TextQA. This could be explained by the text embeddings being segregated in their own vector store without any of the tables; since the answer is certainly contained within that store, the chance of returning the correct chunk is higher, with fewer to choose from.

Despite this potential explanation, the high score on TextQA is surprising, especially when compared with the pure text chunk implementation. For TextQA with the Dual Vector Store method the information returned

by the image vector store is guaranteed to be noise, and vice versa. This contradicts the results of others [10] showing that additional noise reduces performance. This might suggest that noise in the context passed to an LLM might not be as negatively impactful on very direct questions with specific answers contained in the context. However, on the other hand, since the majority of answers can only be found in one chunk, it could be argued that all the methods contain equal amounts of noise.

Overall, we demonstrate that when spatial reasoning is required on table-based questions, the use of image-based encoding gives the best results and that a fine-tuned embedding model will give the best retrieval performance. We show that pure image implementation exhibits the best performance but a dual image and text vector store implementation is also viable. We can also see that the addition of CoT can enhance the reasoning of an LLM on table-based questions in any modality. With the combination of these techniques we are able to achieve our objective of closing the gap between text and table questions, ultimately achieving a higher performance on TableQA than TextQA with the best implementation.

4 Conclusion

This research successfully addresses a critical gap in RAG system performance when processing technical documents containing reference tables. Our experimental results demonstrate that standard RAG implementations exhibit a substantial 24.4 percentage point performance degradation on table-based questions compared to text-based, confirming the practical significance of this problem for organisations deploying RAG systems on their document corpora.

The core contributions of this work lie in demonstrating that image-based approaches outperform text-based methods for table retrieval in RAG contexts and that CoT can be effectively leveraged for table comprehension. Our best-performing implementation, that combines ColPali’s per-page image embedding with multi modal Chain-of-Thought reasoning, achieved a 35.5 percentage point improvement on table-based questions (from 48.9% to 84.4%) while maintaining competitive performance on text queries. This represents the first comprehensive end-to-end evaluation of table-specific RAG performance. Additionally, the success of our novel Dual Vector Store framework balances performance with practicality and flexibility of implementation. This establishes a clear path toward production-ready implementations mixing various text sources with reference tables allowing for modality specific optimisations.

4.1 Ethical Considerations

Despite these successes, the deployment of RAG systems in production environments raises some legal concerns that organisations must carefully address. In terms of compliance, organisations must ensure that training data and document corpora comply with copyright and licensing restrictions, while regulated industries require robust audit trails and explainability mechanisms to meet sector-specific requirements. The automated generation of responses from technical documentation creates potential liability issues if incorrect interpretations of tabular data lead to operational failures or safety incidents. From a social perspective, while these systems could democratise access to specialised technical knowledge, they also risk creating over-reliance on AI-generated interpretations among non-expert users, potentially leading to poor decision-making and reduced development of human expertise within organisations.

Our research was inspired by the increasing reliance on these systems and an urgency to improve performance on critical tabular data. The achievement of our stated aims does not imply that a RAG system with our architecture could be used uncritically without validation and cross checking of information. Engineers and data scientists deploying these systems must implement comprehensive validation frameworks and maintain awareness of system limitations, particularly given the black-box nature of LLMs processing safety-critical or financially sensitive data. The apparent objectivity of technical table data masks potential historical biases embedded in organisational processes and data collection practices, which RAG systems could inadvertently amplify without careful consideration of data provenance. Additionally, the computational requirements of LLMs raise environmental sustainability concerns, requiring organisations to balance performance benefits against energy consumption. The complexity of these systems also creates transparency challenges, making it difficult for users to understand how answers are generated from tabular sources, a particular concern in high-stakes applications where stakeholders need to verify reasoning processes.

While this research demonstrates significant technical advances in table-based RAG performance, successful deployment requires careful consideration of these broader implications. The intersection of enhanced AI capabilities with organisational knowledge management demands thoughtful approaches to validation, oversight, and ethical deployment that extend well beyond technical performance metrics.

4.2 Future Work

While the ColPali per-page implementation achieved the best performance on our stated objective, the Dual Vector Store approach proved equally effective at closing the text-table gap while offering superior flexibility. By allowing text chunks and tables (or any image) to be processed through separate stores, this architecture creates opportunities for implementing modality-specific optimisations tailored to each content type.

The scope of this dual approach could be expanded to encompass other complex visual structures including diagrams, charts, and technical illustrations. Although our research initially critiqued monolithic approaches to multi modal content, extending the Dual Vector Store architecture across a broader range of modalities represents a natural evolution that could benefit from the lessons learned in table-specific processing.

Current retrieval strategies present significant optimisation opportunities beyond our simplified approach of retrieving the top 2 chunks per vector store. Future investigations should explore dynamic retrieval mechanisms including adaptive top-k values, re-ranking algorithms, and score-based thresholds that could substantially improve system performance by better matching retrieval depth to query complexity. Additionally, fine-tuning ColPali on the individual image/table representation as opposed to the whole PDF page could achieve even further performance gains.

The demonstrated success of Chain-of-Thought prompting opens avenues for more sophisticated reasoning frameworks that combine symbolic and neural approaches. Future research could integrate table parsing libraries and structured query languages with LLMs, potentially creating hybrid systems that leverage both the spatial understanding of vision models and the logical precision of symbolic reasoning.

Finally, the field requires standardised benchmarks specifically designed for comprehensive table-heavy RAG evaluation. Unlike existing datasets that focus primarily on retrieval metrics, future benchmarks should provide complete question-answer pairs alongside query-table relationships, spanning diverse domains and table structures to enable complete system evaluation and development.

4.3 Personal Reflection & Lessons Learned

This project provided substantial learning opportunities across multiple dimensions of AI research. Initially, my approach was overly focussed on implementation and dataset creation, leading to significant time investment in baseline systems before conducting a thorough literature review. This resulted in rediscovering established techniques rather than building upon state-of-the-art foundations. A great example of this is the late discovery of ColPali and other VLMs which required substantial architectural pivots that could have been avoided with more comprehensive initial research.

Furthermore, technical implementation revealed the practical complexities of multimodal RAG systems. Building custom APIs for ColPali inference and developing LanceDB interfaces provided valuable hands-on experience with production-level ML infrastructure. However, these technical challenges consumed significant time that could have been used for experimental work if discovered and handled sooner.

The dataset creation process proved more complex and time-intensive than anticipated. Developing the semi-synthetic QA generation utility also came later than ideal in the overall project as I spent the first month reading through documents and writing questions by hand. In future research I will apply this learning and be more streamlined with dataset creation or be more thorough in dataset discovery.

The evaluation framework design process taught important lessons about designing a dataset. Following a more typical dataset design where the image or chunk was saved alongside the question would have made document retrieval scores accessible for all implementations. However, although this was an oversight, the decision to forgo this metric on methods for which it was not applicable was a pragmatic one that saved time.

Overall, I'm pleased with the results I was able to produce and have gained an extensive view of the current state-of-the-art through exploring this research space. I was able to carve out a small niche and demonstrate huge performance gains over the baseline. I came to this project as a relative RAG newcomer and an total research newcomer and to have produced a successful report with novel ideas implementations is a good achievement.

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AI was used in a functional/assistive role in this project, helping to debug code issues and understand concepts.
Tool Used: ChatGPT, OpenAI - <https://chatgpt.com/>

Appendix

LLM-as-a-Judge Prompt:

You are a response checker and your task is to look at two answers to a question. One of the answers is predefined and correct, the other is an LLMs answer.

Respond with a Y or N for the correctness of the answer and a similarity score 1-5. 1 means the answers are completely different and 5 means they are exactly the same in meaning.

Question Generation Prompt:

You are an expert Quality Assurance Analyst with 20 years' experience. Your task is to write a question to evaluate a RAG model based on the text provided. You are not trying to trick the model so the question should be very clear. The answer to each question should be 1-5 words and not include anything other than the succinct information.

Make your questions something that can only be answered by this section of text. Your question should NEVER be about parts of the document such as sections, appendices or paragraphs. Imagine the model has access to many such pages and documents so the question should be only about the content that is answerable in the context of a large corpus.

Basic Answering Prompt:

You are a helpful assistant. You will answer the users question using the context provided. Please use only the context provided.

The question will always have a short answer. Please answer with only the direct 1-5 word answer without rephrasing the question or providing extra context.

Chain-of-Thought Answering Prompt:

Q: What is the 25th percentile chemistry input to compare for Sodium

A: The 25th percentile is on row 3. Sodium has symbol Na so looking down the column for Na to row 3 gives the answer 22.4. So the final answer is 22.4

OCR Prompt:

You are an image parser. Your task is to take in an image of a PDF page and transcribe the text. Your output should only include the text and never any tables or diagrams. Only the paragraphs of text.

If there is no text to parse. Just respond with "No Text"

Table Captioning Prompt:

You are a specialist in table description. You will be given an image of a table or diagram and your task is to return a one-sentence description of that image.

The description should be a clear description of what the table contains without mentioning explicit values.

Don't mention the table or the image, just the information.

Example: Metal sulfides along with their logarithmic equilibrium constants from two different sources.

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